

PART A — MODULE 1 QUESTIONS

MATH4-M1-001

Prompt:

What is the solution to $9 - \frac{x}{2} = 4$?

- A) 5
- B) 8
- C) 10
- D) 13

MATH4-M1-002

Prompt:

A recipe for fruit punch uses 3 cups of juice for every 2 cups of water. If a pitcher contains 20 cups of fruit punch made using this recipe, how many cups of juice are in the pitcher?

- A) 8
- B) 10
- C) 12
- D) 15

MATH4-M1-003

Prompt:

The numbers of minutes a student practiced piano on 7 days are 20, 25, 30, 30, 35, 40, and 45. What is the median number of minutes?

- A) 25
- B) 30
- C) 35
- D) 40

MATH4-M1-004

Prompt:

A water tank contains 45 gallons of water. Water drains from the tank at a constant rate of 3 gallons per minute. Which equation gives the number of gallons G remaining after t minutes?

- A) $G = 45 + 3t$
- B) $G = 45 - 3t$
- C) $G = 3t - 45$
- D) $G = 45t - 3$

MATH4-M1-005

Prompt:

A museum ticket costs \$7, and each audio guide costs \$5. A visitor buys one ticket and g audio guides. If the total cost is less than \$30, which inequality represents this situation?

- A) $7g + 5 < 30$
- B) $5g + 7 < 30$
- C) $5g - 7 < 30$
- D) $7g - 5 < 30$

MATH4-M1-006

Prompt:

What is the value of y in the solution to the system of equations?

$$y = 2x - 1$$

$$x + y = 14$$

- A) 5
- B) 7
- C) 9
- D) 11

MATH4-M1-007

Prompt:

Which expression is equivalent to $2(3r - 5) + 4r$?

- A) $6r - 10$
- B) $10r - 10$
- C) $10r - 5$
- D) $14r - 10$

MATH4-M1-008

Prompt:

Which expression is equivalent to $(a^2)(a^5)$?

- A) a^3
- B) a^7
- C) a^{10}
- D) $2a^7$

MATH4-M1-009

Prompt:

An integer from 1 through 12, inclusive, is chosen at random. What is the probability that the integer chosen is a multiple of 3?

- A) $1/6$
- B) $1/4$
- C) $1/3$
- D) $1/2$

MATH4-M1-010

Prompt:

The area of a square is 49 square centimeters. What is the perimeter, in centimeters, of the square?

- A) 14
- B) 21
- C) 28
- D) 49

MATH4-M1-011

Prompt:

The coordinates of point A are (2, 9), and the coordinates of point B are (10, 9). What is the x-coordinate of the midpoint of segment AB?

- A) 4
- B) 5
- C) 6
- D) 8

MATH4-M1-012

Prompt:

What is the value of $4(6 - 2)^2$?

- A) 16

- B) 32
- C) 48
- D) 64

MATH4-M1-013

Prompt:

If $m(x) = 3x + 8$, what is $m(6)$?

- A) 17
- B) 18
- C) 24
- D) 26

MATH4-M1-014

Prompt:

A bookshelf has 4 shelves. Each shelf holds the same number of books. If the bookshelf holds 72 books total, how many books does each shelf hold?

- A) 12
- B) 16
- C) 18
- D) 24

MATH4-M1-015

Prompt:

A jacket is marked down from \$60 to \$45. What is the dollar amount of the markdown?

- A) \$10
- B) \$15
- C) \$20
- D) \$25

MATH4-M1-016

Prompt:

The table shows the relationship between x and y .

x : 1, 2, 3, 4

y : 4, 8, 12, 16

Which equation represents this relationship?

- A) $y = x + 3$
- B) $y = 2x + 2$
- C) $y = 4x$
- D) $y = 16x$

MATH4-M1-017

Prompt:

Two lines intersect to form vertical angles. One of the angles measures 58° . What is the measure of the vertical angle to it?

- A) 29°
- B) 58°
- C) 122°
- D) 180°

MATH4-M1-018

Prompt:

What is the value of $\sqrt{144} \div 3$?

- A) 3
- B) 4
- C) 6
- D) 12

MATH4-M1-019

Prompt:

A jar contains 12 white beads and 8 black beads. What percent of the beads in the jar are black?

- A) 20%
- B) 30%
- C) 40%
- D) 60%

MATH4-M1-020

Prompt:

The equation $d = 55t$ gives the distance d , in miles, that a car travels in t hours at a constant speed. What is the meaning of 55 in this equation?

- A) The number of hours traveled
- B) The speed of the car, in miles per hour
- C) The total distance traveled after 55 hours
- D) The starting distance, in miles

MATH4-M1-021

Prompt:

A rectangular field has a perimeter of 50 meters. If the length of the field is 15 meters, what is the width, in meters?

- A) 5
- B) 10
- C) 20
- D) 35

MATH4-M1-022

Prompt:

The table shows the numbers of students who signed up for three after-school activities.

Activity: Coding, Chess, Robotics

Number of students: 18, 12, 10

What fraction of the students signed up for chess?

- A) $\frac{1}{4}$
- B) $\frac{3}{10}$
- C) $\frac{2}{5}$
- D) $\frac{3}{4}$

PART B — MODULE 2 QUESTIONS

MATH4-M2-001

Prompt:

What is the solution to $3(4x - 7) - 5x = 28$?

- A) 5
- B) 6
- C) 7
- D) 9

MATH4-M2-002

Prompt:

The system of equations has solution (x, y) .

$$3x + 2y = 32$$

$$x + 2y = 16$$

What is the value of xy ?

- A) 24
- B) 32
- C) 48
- D) 64

MATH4-M2-003

Prompt:

The function f is defined by $f(x) = (x + 1)^2 - 9$. The graph of $y = f(x)$ has two x -intercepts. What is the smaller x -intercept?

- A) -4
- B) -3
- C) 2
- D) 8

MATH4-M2-004

Prompt:

For $x \neq -3$, which expression is equivalent to $(x^2 + 7x + 12)/(x + 3)$?

- A) $x + 3$
- B) $x + 4$
- C) $x + 7$
- D) $x + 12$

MATH4-M2-005

Prompt:

The function p is defined by $p(x) = 2x^2 - 8x + 5$. Which statement about the graph of $y = p(x)$ is true?

- A) The graph has a minimum value of -3 .
- B) The graph has a minimum value of 5.
- C) The graph has a maximum value of -3 .
- D) The graph has a maximum value of 5.

MATH4-M2-006

Prompt:

A savings account balance is \$500 at the beginning of a year. At the end of the year, the balance has increased by 6%. How much money, in dollars, is in the account at the end of the year?

- A) \$506
- B) \$530
- C) \$560
- D) \$600

MATH4-M2-007

Prompt:

The table shows the number of pages remaining in a printer tray after different numbers of print jobs.

Print jobs: 0, 2, 5, 7

Pages remaining: 180, 156, 120, 96

The relationship is linear. How many pages remain after 10 print jobs?

- A) 48
- B) 60
- C) 72
- D) 84

MATH4-M2-008

Prompt:

An isosceles triangle has two equal sides of length 13 units and a base of length 10 units. What is the height of the triangle to the base?

- A) 5
- B) 10
- C) 12
- D) 15

MATH4-M2-009

Prompt:

In a right triangle, $\sin \theta = 8/17$. Which of the following could be the value of $\cos \theta$?

- A) $8/15$
- B) $8/17$
- C) $15/17$
- D) $17/15$

MATH4-M2-010

Prompt:

Line k is represented by $y = (3/4)x - 6$. A line parallel to line k passes through the point $(8, 5)$. What is the y -intercept of the parallel line?

- A) -1
- B) 1
- C) 5
- D) 11

MATH4-M2-011

Prompt:

The functions f and g are defined by $f(x) = x - 4$ and $g(x) = x^2 + 2$. What is $g(f(7))$?

- A) 5
- B) 9
- C) 11
- D) 25

MATH4-M2-012

Prompt:

What is the solution to $2\sqrt{x - 3} = 6$?

- A) 6
- B) 9
- C) 12
- D) 15

MATH4-M2-013

Prompt:

A circle in the xy -plane has diameter with endpoints $(-2, 1)$ and $(4, 1)$. What is the area, in square units, of the circle?

- A) 3π
- B) 6π
- C) 9π
- D) 36π

MATH4-M2-014

Prompt:

For what value of b does the equation $2x + b = 2(x + 5)$ have infinitely many solutions?

- A) 2
- B) 5
- C) 10
- D) 20

MATH4-M2-015

Prompt:

The table shows values of a function q .

x : $-1, 0, 1, 2$
 $q(x)$: $7, 2, -1, -2$

Which equation defines q ?

- A) $q(x) = x^2 - 4x + 2$
- B) $q(x) = x^2 - 2x + 2$
- C) $q(x) = -x^2 - 4x + 2$
- D) $q(x) = -x^2 - 2x + 2$

MATH4-M2-016

Prompt:

Which expression is equivalent to $(18x^4)/(6x^2)$, where $x \neq 0$?

- A) $3x^2$
- B) $3x^6$
- C) $12x^2$
- D) $12x^6$

MATH4-M2-017

Prompt:

A school committee has 6 juniors and 9 seniors. If 5 of the seniors are in the band, what fraction of the committee members are seniors who are in the band?

- A) $1/5$
- B) $1/3$
- C) $5/9$
- D) $5/6$

MATH4-M2-018

Prompt:

A prism has a constant cross-sectional area of 18 square centimeters and a height of 11 centimeters. What is the volume, in cubic centimeters, of the prism?

- A) 58
- B) 99
- C) 198
- D) 396

MATH4-M2-019

Prompt:

For all values of x , $4x^2 - 20x + c = 4(x - 2.5)^2 + 1$. What is the value of c ?

- A) 16
- B) 24
- C) 25
- D) 26

MATH4-M2-020

Prompt:

The values 8, 12, 14, and 18 are in a data set. A fifth value is added so that the mean of the 5 values is 15. What is the fifth value?

- A) 15
- B) 18
- C) 21
- D) 23

MATH4-M2-021

Prompt:

A store sells notebooks for \$3 each and folders for \$2 each. A student buys a total of 11 notebooks and folders for \$28. How many notebooks does the student buy?

- A) 4
- B) 5
- C) 6
- D) 7

MATH4-M2-022

Prompt:

The graph of a linear function contains the points (2, 11) and (8, 35). If the function is written as $f(x) = mx + b$, what is the value of $m + b$?

- A) 1
- B) 3
- C) 5
- D) 7

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH4-M1-001 — Correct: 10 — Explanation: Subtract 9 from both sides to get $-x/2 = -5$.

Multiplying both sides by -2 gives $x = 10$.

MATH4-M1-002 — Correct: 12 — Explanation: The total ratio parts are $3 + 2 = 5$. Juice is $3/5$ of the punch, so the amount of juice is $(3/5)(20) = 12$ cups.

MATH4-M1-003 — Correct: 30 — Explanation: The data are already ordered. With 7 values, the

median is the 4th value, which is 30.

MATH4-M1-004 — Correct: $G = 45 - 3t$ — Explanation: The tank starts with 45 gallons, and 3 gallons are removed each minute, so the amount remaining is $45 - 3t$.

MATH4-M1-005 — Correct: $5g + 7 < 30$ — Explanation: The total cost is 7 dollars plus 5 dollars for each of g audio guides, and this total is less than 30.

MATH4-M1-006 — Correct: 9 — Explanation: Substitute $y = 2x - 1$ into $x + y = 14$ to get $x + 2x - 1 = 14$. Then $3x = 15$, so $x = 5$ and $y = 9$.

MATH4-M1-007 — Correct: $10r - 10$ — Explanation: Distribute first: $2(3r - 5) + 4r = 6r - 10 + 4r = 10r - 10$.

MATH4-M1-008 — Correct: a^7 — Explanation: When multiplying powers with the same base, add exponents: $(a^2)(a^5) = a^7$.

MATH4-M1-009 — Correct: $1/3$ — Explanation: The multiples of 3 from 1 through 12 are 3, 6, 9, and 12. There are 4 favorable outcomes out of 12, so the probability is $4/12 = 1/3$.

MATH4-M1-010 — Correct: 28 — Explanation: A square with area 49 has side length $\sqrt{49} = 7$. Its perimeter is $4(7) = 28$.

MATH4-M1-011 — Correct: 6 — Explanation: The x -coordinate of the midpoint is the average of the x -coordinates: $(2 + 10)/2 = 6$.

MATH4-M1-012 — Correct: 64 — Explanation: Evaluate inside the parentheses first: $6 - 2 = 4$. Then $4(4^2) = 4(16) = 64$.

MATH4-M1-013 — Correct: 26 — Explanation: Substitute 6 for x : $m(6) = 3(6) + 8 = 18 + 8 = 26$.

MATH4-M1-014 — Correct: 18 — Explanation: Divide the total number of books by the number of shelves: $72 \div 4 = 18$.

MATH4-M1-015 — Correct: \$15 — Explanation: The markdown is the original price minus the sale price: $60 - 45 = 15$ dollars.

MATH4-M1-016 — Correct: $y = 4x$ — Explanation: Each y -value is 4 times the corresponding x -value, so the equation is $y = 4x$.

MATH4-M1-017 — Correct: 58° — Explanation: Vertical angles have equal measures, so the vertical angle also measures 58° .

MATH4-M1-018 — Correct: 4 — Explanation: $\sqrt{144} = 12$, and $12 \div 3 = 4$.

MATH4-M1-019 — Correct: 40% — Explanation: There are 20 beads total and 8 are black. Since $8/20 = 0.40$, 40% of the beads are black.

MATH4-M1-020 — Correct: The speed of the car, in miles per hour — Explanation: In $d = 55t$, the coefficient 55 represents the number of miles traveled per hour.

MATH4-M1-021 — Correct: 10 — Explanation: The perimeter is $2(\text{length} + \text{width})$, so $50 = 2(15 + w)$. Then $25 = 15 + w$, so $w = 10$.

MATH4-M1-022 — Correct: $3/10$ — Explanation: There are $18 + 12 + 10 = 40$ students total, and 12 signed up for chess. The fraction is $12/40 = 3/10$.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH4-M2-001 — Correct: 7 — Explanation: Expand to get $12x - 21 - 5x = 28$. Then $7x - 21 = 28$, so $7x = 49$ and $x = 7$.

MATH4-M2-002 — Correct: 48 — Explanation: Subtract the second equation from the first to get $2x = 16$, so $x = 8$. Then $8 + 2y = 16$, so $y = 4$. Therefore, $xy = 32$.

MATH4-M2-003 — Correct: -4 — Explanation: Set $(x + 1)^2 - 9 = 0$, so $(x + 1)^2 = 9$. Then $x + 1 = \pm 3$, giving $x = -4$ or $x = 2$. The smaller x -intercept is -4.

MATH4-M2-004 — Correct: $x + 4$ — Explanation: Factor the numerator: $x^2 + 7x + 12 = (x + 3)(x + 4)$. Since $x \neq -3$, the expression simplifies to $x + 4$.

MATH4-M2-005 — Correct: The graph has a minimum value of -3. — Explanation: Complete the square: $2x^2 - 8x + 5 = 2(x - 2)^2 - 3$. Since the coefficient of the squared term is positive, the graph has a minimum value of -3.

MATH4-M2-006 — Correct: \$530 — Explanation: A 6% increase on \$500 is $0.06 \times 500 = \$30$, so the ending balance is \$530.

MATH4-M2-007 — Correct: 60 — Explanation: From 0 to 2 print jobs, pages decrease by 24, so the rate is -12 pages per print job. After 10 print jobs, pages remaining are $180 - 12(10) = 60$.

MATH4-M2-008 — Correct: 12 — Explanation: The height bisects the base, creating a right triangle with hypotenuse 13 and one leg 5. The height is $\sqrt{13^2 - 5^2} = \sqrt{144} = 12$.

MATH4-M2-009 — Correct: 15/17 — Explanation: In a right triangle with $\sin \theta = 8/17$, the opposite side can be 8 and the hypotenuse 17. The adjacent side is $\sqrt{17^2 - 8^2} = 15$, so $\cos \theta = 15/17$.

MATH4-M2-010 — Correct: -1 — Explanation: A parallel line has the same slope, $3/4$. Using $y = (3/4)x + b$ and the point $(8, 5)$, $5 = 6 + b$, so $b = -1$.

MATH4-M2-011 — Correct: 11 — Explanation: First find $f(7) = 7 - 4 = 3$. Then $g(3) = 3^2 + 2 = 11$.

MATH4-M2-012 — Correct: 12 — Explanation: Divide both sides by 2 to get $\sqrt{x - 3} = 3$. Squaring gives $x - 3 = 9$, so $x = 12$.

MATH4-M2-013 — Correct: 9π — Explanation: The diameter from $(-2, 1)$ to $(4, 1)$ has length 6, so the radius is 3. The area is $\pi(3^2) = 9\pi$.

MATH4-M2-014 — Correct: 10 — Explanation: The right side expands to $2x + 10$. For the equation $2x + b = 2x + 10$ to have infinitely many solutions, b must equal 10.

MATH4-M2-015 — Correct: $q(x) = x^2 - 4x + 2$ — Explanation: Substituting $x = -1, 0, 1$, and 2 into $x^2 - 4x + 2$ gives 7, 2, -1 , and -2 , matching the table.

MATH4-M2-016 — Correct: $3x^2$ — Explanation: Divide coefficients and subtract exponents: $(18x^4)/(6x^2) = 3x^2$.

MATH4-M2-017 — Correct: $1/3$ — Explanation: There are $6 + 9 = 15$ committee members total, and 5 are seniors in the band. The fraction is $5/15 = 1/3$.

MATH4-M2-018 — Correct: 198 — Explanation: The volume of a prism is cross-sectional area times height, so the volume is $18 \times 11 = 198$.

MATH4-M2-019 — Correct: 26 — Explanation: Expand the right side: $4(x - 2.5)^2 + 1 = 4(x^2 - 5x + 6.25) + 1 = 4x^2 - 20x + 25 + 1 = 4x^2 - 20x + 26$. Thus $c = 26$.

MATH4-M2-020 — Correct: 23 — Explanation: A mean of 15 for 5 values means the total is 75. The four given values have sum $8 + 12 + 14 + 18 = 52$, so the fifth value is $75 - 52 = 23$.

MATH4-M2-021 — Correct: 6 — Explanation: Let n be notebooks and f be folders. Then $n + f = 11$ and $3n + 2f = 28$. Substitute $f = 11 - n$ to get $3n + 2(11 - n) = 28$, so $n + 22 = 28$ and $n = 6$.

MATH4-M2-022 — Correct: 7 — Explanation: The slope is $m = (35 - 11)/(8 - 2) = 24/6 = 4$. Using $f(x) = 4x + b$ and point $(2, 11)$, $11 = 8 + b$, so $b = 3$. Therefore, $m + b = 7$.